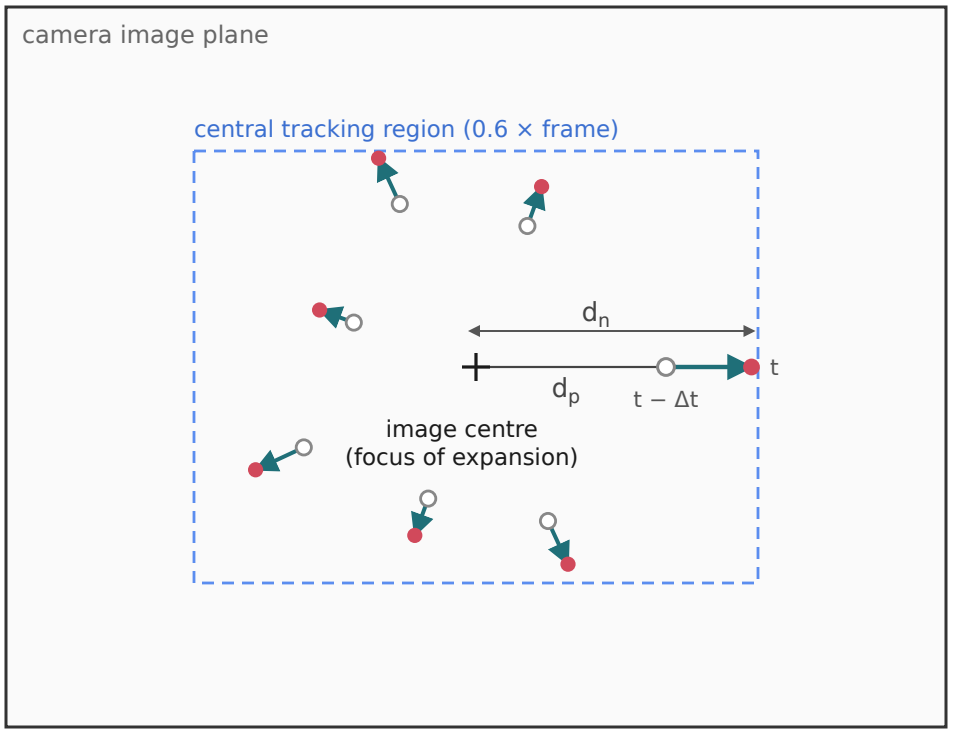
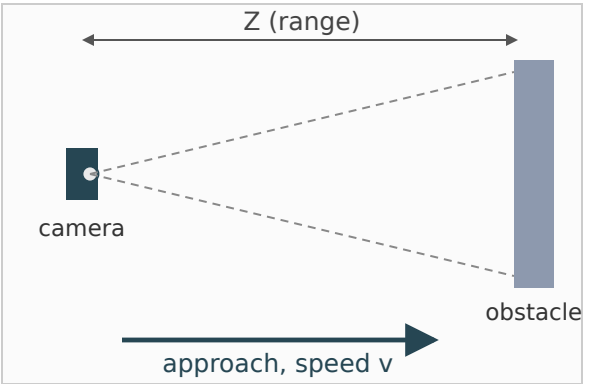
Looming detection by radial optical-flow expansion

(a)



○ feature at t − Δt ● feature at t → optical flow (radial) - - - tracking region

(b)



Time-to-contact (TTC)

$\tau \approx \Delta t / (r - 1)$

$r = \text{median}(d_n / d_p)$, $r > 1 \Rightarrow$ approaching

expansion rate $\propto v / Z$

alert when $\tau < \tau_th$ for N frames

(τ_th = 1.5 s, N = 3 consecutive frames)